PeakMeister 2.0

GUI Guide — Sidebar, Panels, Inputs and Outputs

Based on ui.R and server.R at code commit 709b1bb0997c28519ed8eb144c43ffe62e553f7b, and the supplied Downstream Processing screenshot. Updated 19 October 2026.

ui.R 24–55: main sidebar | *server.R: full implementation*

This guide describes the seven sidebar pages and the controls, tabs and dialogs within them. Inputs include uploaded files, selected rows, typed values and plot gestures. Outputs include visible tables/plots, reactive data in memory, and files written to disk. The task-boundary PDF provides the corresponding server line ranges for every listed task.

How to read your screenshot

Area	What it controls
Far-left dark sidebar	Chooses the main page. Downstream Processing is selected. These are the seven main pages, not analysis stages.
Middle control column within Downstream Processing	Shared controls for loading results, file/peak selection, missing values, normalization, control charts and metadata connection. These remain visible while switching the right-hand tabs.
Right-hand tabbed workspace	m/z vs Migration Time; Control Charts; Data Quality & Variability; Metadata; Batch Correction. The screenshot has Data Quality & Variability selected.
Inside the selected Data Quality & Variability tab	Left: QC identifier, exclusions, Calculate CVs and Average CVs. Right: PCA output. Below: CV-versus-m/z output. Blank regions alone do not establish whether analysis has run.

A GUI action is linked to a task through its Shiny ID. For example, Calculate CVs sends input$compute_cv to server.R 5718–5809. That handler requests grouping/statistics and populates average_cv_table, pca_plot and cv_plot. ui.R 479, 482 and 488 place those outputs in the selected tab. Some reactive outputs recompute when their underlying state changes, without a separate button.

Main sidebar map

Sidebar item / internal tab name	Panels and main result
About / about	Disclaimer, README, Updates, License. Documentation only.
User Supplied Parameters / userparameters	Mass List; Reference Mass List; Parameters; Project Information; User Supplied Migration Indexes. Shared Save Table download.
Engine / engine	Real mz5 upload, pseudo-file-name controls, Initialize Run. Produces Results files through background processing stages.
Visualization / processing	Shared Results/file/plot selection. Plotting and Dual EIE View tabs; peak editing, database migration, baseline and Other dialogs.
Downstream Processing / datamanipulation	Shared result-loading/transformation controls plus five right-hand tabs. Operates mainly on exported result matrices.
Reporting / reporting	Class-assignment form and MetaboAnalyst Matrix output. Uses Downstream Processing's current data/folder.
Drug Analysis / drugs	Adherence Excel upload and drug database loading/parsing. No completed adherence/interaction analysis panel yet.

About

Documentation panels

ui.R 109–129 | *server.R 31–42*

Input / action / output slot	Effect, output or dependency
Disclaimer (expand/collapse)	Displays docs/DISCLAIMER.md through disclaimerContent.
README (expanded initially)	Renders docs/README.md and displays it through readmeContent.
Updates (expand/collapse)	Displays docs/UPDATES.md through updatesContent.
License (expand/collapse)	Displays docs/LICENSE.md through licenseContent.

Inputs are panel expansion/collapse actions; no analytical data are entered and no result files are written here.

User Supplied Parameters

Mass List

ui.R 140–172 | *server.R 49–62, 111–137, 141–206*

Input / action / output slot	Effect, output or dependency
Upload Mass List and Parameters (Excel) [massList]	Reads the Mass List, Reference Mass List, and Parameters worksheets together; updates all three in-memory datasets.
Name; mz; Extraction Window (ppm)	Defines target identity, target m/z and extraction tolerance [name, mz, extraction.window.ppm].
Interference; Interference Comigration Threshold (seconds)	Defines interfering target names and time-overlap threshold [interference, interference.comigration.threshold.seconds].
Minimum Peak Width (seconds); Migration Window (seconds)	Controls candidate width and assignment tolerance [minimim.peak.width.seconds, migration.window.seconds].
Peak Space Tolerance (%); SNR Threshold	Controls spacing checks and the implemented noise/LOD annotation threshold [peak.space.tolerance.percent, snr.threshold].
Smoothing Kernel; Smoothing Strength	Defines optional smoothing settings [smoothing.kernel, smoothing.strength].
Peak 1 MT through Peak 13 MT	Supplies migration-time values for index calculation [peak.1.mt … peak.13.mt].
Add Mass; Delete Selected Row; Clear All	Changes massData; updated Mass List table is shown [addMassRow, deletemassrow, clearMassList → massListData].

The displayed table is marked editable, but no massListData_cell_edit handler is present. Direct cell changes are not established as persistent dataset updates. See the wiring notes at the end.

Reference Mass List

ui.R 175–206 | *server.R 65–77, 111–137, 209–273*

Input / action / output slot	Effect, output or dependency
Upload Mass List and Parameters (Excel) [massList]	Uses the same three-worksheet upload handler as Mass List.
Name of Reference Mass; mz (reference mass); Extraction Window (ppm)	Reference target identity and extraction settings [name, mz, extraction.window.ppm].
Class; Min MT Min; Max MT Min	Reference classification and allowed migration-time limits in minutes [class, min.mt.min, max.mt.min].
Peak FWHM Tolerance Multiplier; Peak Space Tolerance (%)	Reference peak-width and spacing-filter settings [peak.fwhm.tolerance.multiplier, peak.space.tolerance.percent].
Smoothing Kernel; Smoothing Strength	Reference trace smoothing settings [smoothing.kernel, smoothing.strength].

Input / action / output slot	Effect, output or dependency
Peak 1 MT through Peak 13 MT	Visible fields use ref.peak.1.mt … ref.peak.13.mt. The Add Reference Mass handler instead reads peak.1.mt … peak.13.mt; this mismatch needs correction before relying on manual entry.
Add Reference Mass; Delete Selected Row; Clear All	Changes refMassListData and refreshes the reference table [addRefMassRow, deleterefmassrow, clearRefMassList].

Reference targets are used for peak assignment. The two mass-calibration lock masses are configured separately in Parameters. Editable table cells have no matching persistence handler in this server.

Parameters

ui.R 209–230 | *server.R 80–91, 111–137, 276–320*

Input / action / output slot	Effect, output or dependency
Upload Mass List and Parameters (Excel) [massList]	Reads all three main worksheets.
number.of.injections	Number of injections/expected peak positions per run.
ref.mass.one; ref.mass.two	Lower and upper known m/z references for optional mass calibration.
ref.mass.window.ppm; ref.mass.minimum.counts	Search window and minimum intensity for calibration references.
apply.mass.calibration; apply.smoothing	Yes/No selectors enabling calibration and smoothing.
plot.format	Sample or Metabolite; controls the organization of saved PNG plots.
Manual.Indexes	Yes/No selector enabling user-supplied migration-index overrides.
Add Parameter; Delete Selected Row; Clear All	Changes parametersData and the Parameters table [addParamRow, deleteparamrow, clearparam].

The parameter selectors use Yes/No text values. Editable table cells have no matching persistence handler in the reviewed server.

Project Information

ui.R 233–259 | *server.R 326–358*

Input / action / output slot	Effect, output or dependency
Project Name; Project Description; Project supervisor; Project Contact	Text fields [projectName, projectDescription, projectSupervisor, projectContact].
Save Project Info [saveProjectInfo]	Stores text in reactive state and shows Current Project Info [currentProjectInfo].
Clear All [clearproject]	Clears both entry fields and stored project information.

The reviewed handler does not export a project-information file or use this state to name Results folders. The "REQUIRED" wording is a UI label, not validation implemented by this handler.

User Supplied Migration Indexes

ui.R 262–288 | *server.R 95–101, 363–426; application to run: 743–766*

Input / action / output slot	Effect, output or dependency
User Supplied Migration Indexes (Excel) [userMTI]	Reads worksheet “User Supplied Migration Indexes” into userMTI state and displays userMTIdata.
Name of Metabolite; Left internal standard; Right internal standard; Description	Defines the target and reference/index type [name, left_is, right_is, description].
Peak 1 MT through Peak 13 MT	Visible fields use ref.peak.*. The Add manual MTI row handler reads peak.* instead; the manual values are not wired to the displayed fields correctly.
Add manual MTI row; Delete Selected MTI row; Clear All	Adds/removes rows in userMTI state [addMTIrow, deleteMTIrow, clearMTIlist].

Overrides are used by Initialize Run only when Parameters > Manual.Indexes is Yes. Uploaded values are validated against target/reference names and injection count. Editable table cells lack a matching persistence handler.

Shared download below the tabs

ui.R 291–293 | *server.R 431–445*

Input / action / output slot	Effect, output or dependency
Save Table (.xlsx) [downloadData]	Downloads a dated workbook containing Parameters, Mass List and Reference Mass List worksheets.

This download excludes Project Information and User Supplied Migration Indexes. Initialize Run also writes its own copy of the three-sheet workbook into the Results folder.

Engine

File Uploads for Initializing Data Preprocessing

ui.R 301–310 | *server.R 451–486*

Input / action / output slot	Effect, output or dependency
Upload .mz5 Files [mz5Files]	Accepts multiple real raw files; appends their names/temporary paths to uploadedmz5 and displays fileTable.
Start run automatically after upload [autorun]	When checked, uploading files triggers the Initialize Run button.
Clear Files [clearFiles]	Clears uploadedmz5 and the pseudo-name state. This is not a command to delete existing Results folders.

Generate Pseudo .mz5 Files to Access Stored Datafiles

ui.R 313–318 | *server.R 489–541*

Input / action / output slot	Effect, output or dependency
Upload Excel File of File Names (.xlsx) [excelFile]	Reads a FileName column and displays the editable mz5fileTable.
Table cell edits; Add Empty Row [addRow]	Updates pseudo-file names; the mz5fileTable_cell_edit handler is implemented.
Generate Pseudo .mz5 Files [generateFiles]	Intended to create empty placeholders and populate the uploaded-file list for accessing already processed results.

Pseudo files contain no spectra and are not valid substitutes for raw input to Initialize Run. The current code assigns file.create() return values to FilePath rather than the generated path, so this feature has a concrete implementation issue (server.R 520–523).

Initialize Run and progress

ui.R 323–335 | *server.R 546–2531*

Input / action / output slot	Effect, output or dependency
Initialize Run [initialize]	Uses uploaded real mz5 files plus the supplied mass lists, parameters, and optional indexes. Executes reading, calibration, extraction, smoothing, peak processing, plotting, and export.
Progress / console messages	withProgress displays a file-level progress panel; timing instrumentation updates stage messages. The standalone progressBar UI slot has no output$progressBar renderer.
Saved outputs	New Results… directory: Migration Index Summary.csv; Metabolite Peak Areas.csv; Metabolite Migration Times.csv; dated parameter workbook; results.sqlite; plotly_objects.RData; run_timings.csv; Plots/ PNGs; Data/<file>.mz5/plot_list_data.RData.

Extraction produces eie_df in memory; it is not a separate GUI panel. The processing stages are background steps of this one button. Full dataset processing was not executed for this documentation review.

Visualization

Shared left control column

ui.R 342–357 | *server.R 2571–2615, 3127–3379, 3590–3760, 4538–4540*

Input / action / output slot	Effect, output or dependency
Uploaded .RData file [RDatainput]; Clear uploaded data [resetplotlydata]	Visible in ui.R, but neither input has a server handler in the reviewed version. Use the Results-folder/file selection path for the implemented loader.
Select File [file_selector]	Choices come from uploadedmz5 names. Selecting a file loads its SQLite data and associated .RData traces into editing state.
Select Results Folder [results_folder]	Selects a Results directory; opens results.sqlite or offers to build it from existing Data/*/plot_list_data.RData files.
Loaded Plots [plot_table]	Lists targets for the selected file. Selecting a row chooses the right-hand electropherogram.
Edited plots to be regenerated [modified_peak_plots_table]	Lists plot names marked modified by editing actions.

The Visualization and Downstream Processing folder selectors are separate controls; selecting one does not synchronize the other. Result-folder discovery searches immediate directories under the app working directory whose names contain Results.

Database Not Found / Building Database dialogs

UI created in server.R 3187–3201, 3212–3218 | *server.R 2953–3013, 3161–3256*

Input / action / output slot	Effect, output or dependency
Yes — build database [confirm_migration]	Migrates stored .RData results into a new SQLite database, reports progress and failed files, and connects the result.
No — skip	Dismisses the dialog without establishing the missing database connection.

These are dialogs shown after selecting a Results folder without a database, not permanent sidebar items.

Plotting: plot display and integration boundaries

ui.R 361–368, 398–399 | *server.R 3386–3583*

Input / action / output slot	Effect, output or dependency
Select a Loaded Plots row	Displays selected_plot: trace, peak labels/comments, integration shading, and two boundary lines.
Zoom / double-click reset	Stores zoom state and restores it during redraws; selecting a different target resets it.
Draw Mode [draw_mode]; draw rectangle or drag lines	Switches between zoom and drawing; rectangle edges update red/blue integration boundaries.

Input / action / output slot	Effect, output or dependency
Red Line Position / Blue Line Position	Text outputs red_line_position and blue_line_position; plot coordinates are minutes, converted to seconds during integration.
Peak Position Information [peak_info_table]	Shows peak number, comment, apex seconds and height; row selection supplies peaks for edit buttons.

Plotting: peak-edit buttons and dialogs

ui.R 379–396; dialogs created in server | server.R 3763–4056, 4566–5083, 5153–5232

Input / action / output slot	Effect, output or dependency
Delete Selected Peaks [delete_peak]	Marks selected rows NPD; removes corresponding integration shading and persists edits.
Delete Middle Peaks (Keeps first and last) [deletemiddlepeaks]	Marks interior peaks NPD and retains endpoint peaks.
Delete all peaks [deleteallpeaks]	Marks all selected-target peaks NPD and clears their integration shading.
Undo Peak Deletion [undo]	Restores the previous in-memory snapshot and writes the .RData file. This handler does not restore SQLite, so it is not a complete database-backed undo.
Manually Adjust Integration [manual_integrate]	Uses red/blue boundaries and selected peak number; opens Manual Peak Input if no row is selected. Confirmation [confirm_peak_number] integrates and saves the peak.
Adjust Individual Baseline [adjust_indiv_baseline]	Activates plot-click baseline selection. Uses selected peak, or opens a peak-number dialog [baseline_peak_number_adjustment].
Adjust All Baselines [adjust_all_baselines]	Opens a plot dialog [adjust_baseline_plot]; clicking a height recalculates all peak areas. The Confirm button [confirm_adjust_all_baselines] only closes the dialog; recalculation happens on click.
Mark Peak Interference [interferences]	Marks selected peaks interfered and removes their integration shading.

Edits generally flow through save_plot_data(), which updates SQLite, writes an .RData backup, and refreshes the plot. Automatic and manual interference annotations use different spellings in this source. Editing a peak does not rerun raw extraction.

Plotting > Other dialog

ui.R 390; dialog in server.R 4471–4486 | server.R 4161–4187, 4247–4432, 4489–4562

Input / action / output slot	Effect, output or dependency
Regenerate Changed Plots [regenerateplots]	Rebuilds the peak-area CSV, regenerates changed PNGs if mass-list/parameter data are loaded, and clears the modified-plot list.
Rebuild Peak Area CSV [rebuild_peak_areas_modal]	Reads SQLite peak areas and writes Metabolite Peak Areas.csv.
Rebuild Peak Migration Time CSV [rebuild_mig_times_modal]	Reads annotation apex times and writes Metabolite Migration Times.csv.

The saved-Plotly rebuilding helper regenerate_plots() exists but is not called by the regeneration button. Initial migration-time export divides seconds by 60; the rebuild helper writes apex seconds without conversion (server.R 2454–2456 versus 4410–4431).

Dual EIE View

ui.R 402–414 | *server.R 5087–5150*

Input / action / output slot	Effect, output or dependency
Select Plots to Compare: three selection tables [plottable1, plottable2, plottable3]	Choose two or three distinct target rows for the current file.
Combined Electropherograms [combined_plot]	Displays vertically stacked EIE plots with a shared x-axis. No separate export button is defined in this panel.

Downstream Processing

The first five sections below describe the shared control column. The following five describe the right-hand tabs. File selection for m/z plotting does not restrict missing-value replacement, normalization or QC analysis to that file; those actions use the loaded matrix.

Shared controls: Set working environment and initialize data

ui.R 424–429 | *server.R 5237–5278*

Input / action / output slot	Effect, output or dependency
Select Results Folder [results_folder2]	Chooses the downstream Results folder independently of Visualization.
Load Migration and Peak Area Data [load_migration_area]	Reads Metabolite Peak Areas.csv and Metabolite Migration Times.csv into peak_areas_data and migration_times_data. No raw mz5 extraction is performed.

The current peak-area matrix is shared by normalization, missing-data replacement, metadata connection, control charts, QC/PCA, Reporting and batch correction. Loading again reads the original named CSVs; it does not automatically select a previously normalized or merged CSV.

Shared controls: m/z vs MT controls

ui.R 431–435; peak selector created by server | *server.R 5277–5278, 5283–5344*

Input / action / output slot	Effect, output or dependency
Select File [file_selector2]	Chooses which block of migration-time CSV rows to plot; choices come from the Engine uploaded/pseudo-file name list.
Select Peak Number [peak_number]	Chooses the injection/peak position; generated through peak_number_selector from Parameters > number.of.injections.
Right-hand m/z vs Migration Time tab	Displays mz_vs_migration_plot, plotting migration time against m/z with target names and internal-standard/metabolite categories.

This plot also needs the uploaded Mass List and Reference Mass List. The helper derives m/z from target-name prefixes such as 85.1327_TMAO-d9. Raw spectra are not read to generate this plot.

Shared controls: Editing your data matrix

ui.R 437–443; normalization dialog in server.R 5349–5359 | *server.R 5349–5443*

Input / action / output slot	Effect, output or dependency
Select Method for Replacing Missing Data [missing_data_method]	Choose Minimum Values/5 or Missing values = 0.
Replace Missing Data [missing_data]	Converts non-numeric annotations to missing values, applies the selected replacement, updates the current matrix and saves Corrected Missing Values Metabolite Peak Areas.csv.

Input / action / output slot	Effect, output or dependency
Normalize Data [normalize]	Opens Normalization Options. Select a reference target [normalize_by] and press Normalize [confirm_normalize]. Updates the current matrix and saves Normalized Metabolite Peak Areas.csv.
Where to see results	Metadata > Peak Area Datatable shows the current matrix. The processed Control Charts and later analyses use that same matrix.

Normalization reference choices are hardcoded: 114.0667_Creatinine, 184.0774_F-Phe, 216.0427_Cl-Tyr, 182.0617_F-Phe, 198.0566_F-Tyr, 214.0271_Cl-Tyr, 207.0115_NMS and 212.9412_Bromohypoxanthine. Actions operate on the current state, so their order affects later steps.

Shared controls: Generate Control Charts

ui.R 445–447 | *server.R 5514–5575*

Input / action / output slot	Effect, output or dependency
Select Metabolite [selected_metabolite]	Despite the label, choices are populated from Reference Mass List names.
Create Control Chart [create_control_chart]	Triggers both the raw and processed chart handlers. Each chart plots areas by sample with mean and ±3 standard-deviation lines.
Right-hand Control Charts tab	Shows control_chart_raw from the original peak-area CSV, and control_chart_processed from the current in-memory matrix.

The second chart is labelled “normalized data” in the UI, but it reflects whatever processing has been applied to the current matrix. Normalization is not performed automatically by this button.

Shared controls: Connecting metadata

ui.R 449–451 | *server.R 5449–5509*

Input / action / output slot	Effect, output or dependency
Connect Metadata to Results [connect_metadata]	Reads the selected Results folder’s Data/metadata.csv, reshapes injection columns, and matches Data file + peak.number to PBM Sample ID.
Output in memory and on disk	Adds PBM Sample ID to peak_areas_data; Metadata tab displays the result. Writes merged_peak_area.csv in the selected Results folder.

There is no metadata upload widget here. The user must place metadata.csv in Data/. It needs a Data file column and numeric-name injection columns such as 1, 2, 3. Sample IDs are trimmed. QC/PCA and Reporting depend on these IDs.

Right tab: m/z vs Migration Time

ui.R 457–458 | *server.R 5283–5344*

Input / action / output slot	Effect, output or dependency
Inputs come from the shared controls	Loaded migration-time CSV + Mass List/Reference Mass List + Select File + Select Peak Number.
Visible output [mz_vs_migration_plot]	Interactive m/z-versus-migration-time scatter plot. Internal standards and metabolites are coloured separately.

No separate form or save button is defined inside this tab. The plotted x-axis is labelled minutes; the migration-time CSV must use the expected units. See the migration-time rebuild mismatch in the wiring notes.

Right tab: Control Charts

ui.R 460–465 | *server.R 5522–5575*

Input / action / output slot	Effect, output or dependency
Inputs come from Generate Control Charts	Reference-target choice and Create Control Chart button; original CSV for raw chart, current peak-area matrix for processed chart.
Control chart for non-normalized data [control_chart_raw]	Original peak-area values plotted by sample with mean/control limits.
Control chart for normalized data [control_chart_processed]	Current in-memory peak-area values plotted by sample with mean/control limits.

Right tab: Data Quality & Variability (selected in your screenshot)

ui.R 467–491 | *server.R 5582–5809*

Input / action / output slot	Effect, output or dependency
Specify QC identifier (e.g., QC=NIST) [category_input]	Intended to identify QC samples from PBM Sample ID; other samples form the Non-QC group. The matching expression uses case-insensitive grepl(), so its value acts as a pattern.
Exclude PBM Sample IDs [exclude_ids]	Comma-separated exact sample IDs to omit; exclusion matching is case-sensitive.
Calculate CVs [compute_cv]	Requests QC/non-QC grouping, data checks, per-target CVs and PCA.
Average CVs (%) [average_cv_table]	Technical CV is the mean per-target CV among QC samples; biological CV is the mean among non-QC samples.
PCA plot [pca_plot]	Right of the QC form: sample scores for QC and Non-QC, with PC1/PC2 variance labels. Also saves Data/QC_PCA_plot.png.
CV based on m/z of metabolite [cv_plot]	Below the QC form/PCA area: QC and Non-QC CVs by target m/z with median-reference lines.

This panel requires loaded numeric peak areas and connected PBM Sample ID metadata. The screenshot alone does not establish whether Calculate CVs has completed. The QC form and Reporting class form reuse category_input; independent values are not guaranteed by this UI definition.

Right tab: Metadata

ui.R 492–495 | *server.R 5814–5824*

Input / action / output slot	Effect, output or dependency
No extra input form in this tab	Uses the shared/current peak_areas_data matrix.
Peak Area Datatable [peak_area_table]	Shows file/peak information, target areas, replacements/normalization and sample IDs when present. Horizontal/vertical scrolling; first four columns fixed.

This is a view of the current peak-area matrix, not a separate metadata-file editor. Connect Metadata to Results is in the shared control column.

Right tab: Batch Correction

ui.R 497–502 | *server.R 5828–5916*

Input / action / output slot	Effect, output or dependency
Initialize Packages [load_batchcorr_packages]	Installs/loads optional RcppArmadillo, ChemometricsWithR and BatchCorrMetabolomics. May require compiler support and console responses.
Upload Batch Correction Order [batch_corr_order]	Reads a CSV containing SCode, Batch and Seq; SCode = ref selects the reference group. Rows must correspond to the area-matrix sample order.
Perform Batch Correction [batch_corr]	Uses current peak areas, log10 transformation and two correction strategies, then back-transforms results.
Saved outputs; no chart/table defined here	Writes BatchCorrResults.csv and BatchCorrResultsAntilog.csv in the app working directory. The handler does not replace peak_areas_data with the corrected result.

Numeric positive data are needed for the log10 step. This panel is an optional analysis path and does not rerun extraction.

Reporting

Generate Matrix for MetaboAnalyst / MetaboAnalyst Matrix

ui.R 509–523 | *server.R 5920–5990*

Input / action / output slot	Effect, output or dependency
Specify classes for MetaboAnalyst matrix [category_input]	Left control column: comma-separated rules such as SP1=Standard1, NIST=SRM361, Numeric=Pos_Sample. Numeric applies to numeric sample IDs; other rules use case-insensitive pattern matching in this server.
Generate Data Matrix for MetaboAnalyst [generate_matrix]	Uses the downstream current peak-area matrix with PBM Sample ID, removes file/peak columns, transposes, and inserts a CLASS row. Unmatched classes are blank.
MetaboAnalyst Matrix [matrix_table]	Right-hand table displays the generated matrix. Writes Metaboanalyst_matrix.csv in the selected downstream Results folder.

There is no Results-folder selector here: Reporting uses Downstream Processing's main_folder2. This class form shares category_input with QC; independent values are not guaranteed by this UI definition.

Drug Analysis

Data import and DrugBank parsing dialog

ui.R 526–537; dialog in server.R 6030–6038 | *server.R 5999–6092*

Input / action / output slot	Effect, output or dependency
Upload Adherence Results (Excel) [adherence_results]	Reads Excel data into adherence_patient_data. No patient-result table or adherence calculation is wired to this data yet.
Load Drug Databases [drug_databases]	Loads Drug_Data/TWOSIDES.csv.gz and cached Drug_Data/drugbank_db.rds. Databases must be supplied independently.
DrugBank Database Not Found > Yes, parse now [confirm_parse_drugbank]	Parses Drug_Data/full database.xml with selected categories, shows a loading screen, caches drugbank_db.rds and updates drugbank_db state.
Outputs / right-hand region	Notifications and cached database file. No right-hand analysis plot, interaction report, or adherence result panel is defined in the current UI.

The imported databases and patient data are preparation only. The sidebar label does not imply implemented drug-interaction or adherence analysis.

Data flow and implementation notes

How the pages share data

Producer / action	Where the result is consumed
User Supplied Parameters: uploaded/current mass lists, parameters and optional indexes	Engine Initialize Run; plot-regeneration settings; downstream target names, injection count and reference choices.
Engine: uploadedmz5 file-name list	Visualization Select File; Downstream Processing Select File. Pseudo files are intended to populate this list for older results.
Engine: Results…/results.sqlite + Data/<file>.mz5/plot_list_data.RData	Visualization reads traces from .RData and editable annotations/measurements from SQLite. Both belong to the same selected Results folder.
Engine / Visualization: peak-area and migration-time CSVs	Downstream Processing Load Migration and Peak Area Data reads the named CSVs. Changes in Visualization need the relevant CSV rebuilt before reloading downstream.
Downstream: current peak_areas_data matrix	Metadata table, transformations, processed control chart, QC/PCA, batch correction and Reporting. Actions modify/read this shared state; order matters.
Downstream: selected folder main_folder2	Destination for downstream CSVs, QC_PCA_plot.png and Reporting's Metaboanalyst_matrix.csv. Separate from Visualization's folder choice.

Visible controls and source-wiring gaps

Finding	Relevant code / practical meaning
Repeated input IDs across panels	ui.R repeats massList, name, mz, extraction.window.ppm, smoothing controls and reference peak IDs. category_input is reused by QC and Reporting. Distinct-looking controls are not uniquely identified; independent values are not guaranteed.
Reference/manual migration-time field mismatch	ui.R 198 and 279 create ref.peak.* fields. server.R 222–234 and 378–390 read peak.* fields. Manual entries in those panels are not read from their displayed field IDs.
Editable parameter-table cells without write-back handlers	Only mz5fileTable_cell_edit is handled in this server. Mass/reference/parameter/index tables are declared editable, but matching *_cell_edit handlers are absent.
Unimplemented .RData upload/reset path	ui.R 344–346 define RDatainput/resetplotlydata; no corresponding server handlers were found. Implemented result loading uses folder/file selection.
Progress placeholder versus actual progress	ui.R 335 defines progressBar without output$progressBar in server.R. Initialize Run instead uses withProgress and timing stage messages.
Pseudo-file path assignment	server.R 520–523 returns file.create() results as FilePath, rather than the generated temporary paths. Pseudo files also have no raw spectra.
Undo persists only one storage representation	server.R 4012–4056 restores in-memory state and .RData; no matching SQLite restoration appears in that handler. SQLite-backed views can retain edited values.
Migration-time CSV unit inconsistency	Initial export at server.R 2454–2456 divides apex seconds by 60. Rebuild at 4410–4431 writes apex seconds directly. Downstream x-axis labels use minutes.
Some named helpers are not connected to a current action	regenerate_plots(), cv_data, pca_data and plot_list_data have no consuming calls found in this file. The task PDF labels these legacy/uncalled helpers.
Labels versus current state	“Control chart for normalized data” reads the current matrix; it does not normalize it. “Metadata” shows the area matrix. Drug Analysis currently imports data only.

Validation scope

The guide maps visible labels and IDs in ui.R to handlers, reactives and outputs in server.R. It includes dialogs created dynamically by the server. The supplied screenshot was used to clarify layout. No application source was modified and no complete dataset run was performed for this documentation; the wiring findings describe the reviewed source.